

Logic in Computer Science
Problem Sheet 3: Propositional Logic Semantics
August 23, 2025

1. Check whether the valuation v satisfies the *wff*s given below where $v(p_1) = T$, $v(p_2) = F$ and $v(p_3) = T$.
(a) $((p_1 \rightarrow p_2) \rightarrow (\neg p_1))$ (b) $((p_1 \rightarrow p_2) \wedge (p_1 \rightarrow p_3)) \leftrightarrow (p_1 \rightarrow (p_2 \vee p_3))$
2. Check whether the following formulas are valid/satisfiable/unsatisfiable.
(a) $((p_1 \vee p_2) \rightarrow p_1)$ (b) $((p_1 \wedge p_2) \rightarrow p_1)$ (c) $((p_1 \rightarrow p_2) \rightarrow p_1)$
(d) $((p_1 \rightarrow p_2) \rightarrow p_2)$ (e) $((\neg(\neg p_1)) \rightarrow p_1)$ (f) $(p_1 \rightarrow (p_1 \vee p_2))$
3. Prove or disprove:
(a) If $v \models A$ and $v \models (A \rightarrow B)$, then $v \models B$.
(b) If A is satisfiable and $(A \rightarrow B)$ is satisfiable, then B is also satisfiable.
(c) If A is valid and $(A \rightarrow B)$ is valid, then B is also valid.
(d) If a formula A is valid, then A is satisfiable.
(e) If a formula A is satisfiable, then A is valid.
(f) If a formula A is satisfiable, then $(\neg A)$ is unsatisfiable.
4. Prove that a formula A is valid iff $(\neg A)$ is unsatisfiable.
5. Show that the following formulas are valid.
(a) $((p_1 \rightarrow (p_1 \rightarrow p_2)) \rightarrow (p_1 \rightarrow p_2))$
(b) $((p_1 \rightarrow p_3) \rightarrow (p_2 \rightarrow p_3)) \rightarrow ((p_1 \vee p_2) \rightarrow p_3)$
(c) $((p_1 \rightarrow p_3) \rightarrow ((\neg p_1 \rightarrow p_3)) \rightarrow p_3)$
6. *Prove or disprove* the following statements. Given that U_1, U_2 are sets of well formed formulas.
(a) If $U_1 \subseteq U_2$, then $Mod(U_2) \subseteq Mod(U_1)$
(b) If $U_1 \subseteq U_2$, then $Mod(U_1) \subseteq Mod(U_2)$